

An elementary proof of Erdős Problem 488 for sets with at most three primitive generators

Sandbox note (Claude + Codex collaboration)

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Abstract

Let A be a finite set of integers ≥ 2 , let $B = \{k \geq 1 : a \mid k \text{ for some } a \in A\}$ be its set of multiples, and let $B(x) = |B \cap [1, x]|$. Erdős Problem 488 asks whether $B(m)/m < 2 B(n)/n$ for all $m > n \geq \max A$, with the constant 2 known to be best possible. We give a short, self-contained, elementary proof of this inequality whenever the primitive core of A has at most three elements (in particular for every A with $|A| \leq 3$). The engine is the classical two-term Bonferroni bound for a set of multiples—at the density level the Heilbronn–Rohrbach inequality (1937)—kept *finitary* and combined with one observation: each generator’s “charge” is a positive integer. This upgrades the density bound to the sharp constant 2 on the only nontrivial reciprocal regime, via a single one-line parity dichotomy. The argument uses no counting function beyond B , no case split on n , and no transport machinery; we argue it is formalizable in Lean without the case-analysis gap left unproved in the one existing proof of this statement, and we record the exact obstruction to four generators. We claim no novel *method* (the backbone is classical) and no priority for the *result* (it is Chojecki’s, below); the contribution is a correct, elementary, and formally clean proof of a case where the public frontier is stuck.

1 Introduction

For a finite set $A \subset \{2, 3, 4, \dots\}$ write

$$B = B_A = \{k \geq 1 : a \mid k \text{ for some } a \in A\}, \quad B(x) = |B \cap [1, x]|.$$

Erdős [1, Problem 27, p. 236] asked (after correcting a typo in the printed statement; see Remark 13):

(EP488) Is it true that for every $m > n \geq \max A$, $\frac{B(m)}{m} < 2 \frac{B(n)}{n}$?

The constant 2 cannot be lowered: for $A = \{a\}$, $n = 2a - 1$, $m = 2a$ one has $B(n) = 1$, $B(m) = 2$, and the ratio $\frac{B(m)/m}{B(n)/n} = 2 - \frac{1}{a} \rightarrow 2$. This witness is Erdős’s own, and it only makes sense for the multiples reading of the problem, which fixes the intended statement (Remark 13).

The problem is listed as *open* on erdosproblems.com/488 (status unchanged as of July 2026: “no solutions, partial or complete, claimed”). On the discussion thread the case $|A| = 2$ is settled, and the most complete attempt at larger sets is a 27-page note of Chojecki [2] (with a Lean formalization) which *claims* (EP488) for every A whose *primitive core* has at most three elements. That proof proceeds through a signed-lcm “transport” identity, a two-block decomposition, and a “pair-against-one-forbidden-modulus” theorem whose crux is a combinatorial lemma (“there are at least four exact-one points”) that is left as an unverified `sorry` in the accompanying Lean

development; moreover Tao (April 2026) exhibited a family showing the reduction underlying that step *fails* once $\min A \geq 3$. So the three-generator case is, in practice, still open.

Our contribution is a proof of the three-generator statement that is elementary, self-contained, and—we argue—formally clean. We prove the stronger *order-free* bound

$$\sup_{m \geq 1} \frac{B(m)}{m} < 2 \inf_{n \geq \max A} \frac{B(n)}{n}$$

by establishing, in the only nontrivial regime, the pointwise estimate $2B(n) > nS$ for every $n \geq \max A$, where $S = \sum_{d \in P} 1/d$ is the reciprocal sum of the primitive core P (Section 1). The one substantive step is a parity dichotomy (Lemma 6); everything else is the finitary Heilbronn–Rohrbach bound plus an integrality remark. We make no claim of priority for the statement (Chojecki’s Corollary 4.7), and none of novelty for the method (Remark 14); we record only that the proof is different, elementary, and free of the formalization gap.

Notation. A set $\{a, b, c\}$ with $a < b < c$ is *primitive* if no element divides another. Replacing A by its primitive core P (the elements of A divisible by no smaller element of A) leaves B unchanged and satisfies $\max P \leq \max A$, so it suffices to prove (EP488) for primitive A on the wider range $n \geq \max P$; and $S := \sum_{d \in P} 1/d$ is then the relevant reciprocal sum. For a primitive triple $\{a, b, c\}$,

$$S = \frac{1}{a} + \frac{1}{b} + \frac{1}{c}, \quad s(n) = \left\lfloor \frac{n}{a} \right\rfloor + \left\lfloor \frac{n}{b} \right\rfloor + \left\lfloor \frac{n}{c} \right\rfloor, \quad P_2(n) = \left\lfloor \frac{n}{L_{ab}} \right\rfloor + \left\lfloor \frac{n}{L_{ac}} \right\rfloor + \left\lfloor \frac{n}{L_{bc}} \right\rfloor,$$

where $L_{xy} = \text{lcm}(x, y)$. Since $1 \notin B$, we always have $B(x) \leq x - 1$.

2 Two elementary ingredients

Proposition 1 (Singleton case, sharp). *If $A = \{a\}$ with $a \geq 2$, then $B(m)/m < 2B(n)/n$ for all $m > n \geq a$.*

Proof. Here $B(x) = \lfloor x/a \rfloor$, so $B(m)/m \leq 1/a$ for all m . Writing $n = a \lfloor n/a \rfloor + r$ with $0 \leq r < a$ and $\lfloor n/a \rfloor \geq 1$ (as $n \geq a$), we have $a \lfloor n/a \rfloor \geq a > r$, hence $2a \lfloor n/a \rfloor > a \lfloor n/a \rfloor + r = n$, i.e. $2B(n)/n = 2 \lfloor n/a \rfloor / n > 1/a \geq B(m)/m$. $\square$

Proposition 2 (Reciprocal-sparse core: the covered zone). *Let P be a primitive set with least element a and $|P| \geq 2$, and suppose $\sum_{d \in P} 1/d \leq 2/a$. Then (EP488) holds for P .*

Proof. Fix $n \geq \max P$. Pick any $b \in P \setminus \{a\}$; then $b \leq \max P \leq n$, $b \in B$, and $a \nmid b$ by primitivity, so b is not among the $\lfloor n/a \rfloor$ multiples of a in $[1, n]$. Hence $B(n) \geq \lfloor n/a \rfloor + 1 > n/a$. On the other hand $B(m) \leq \sum_{d \in P} \lfloor m/d \rfloor \leq m \sum_{d \in P} 1/d \leq 2m/a$ for every m . Therefore $B(m)/m \leq 2/a < 2B(n)/n$. $\square$

For a primitive triple the hypothesis of Proposition 2 reads $\frac{1}{b} + \frac{1}{c} \leq \frac{1}{a}$. Thus only the *uncovered zone*

$$\frac{1}{b} + \frac{1}{c} > \frac{1}{a} \tag{1}$$

remains; everything in the next two sections concerns that case.

3 Floor facts, ratio bounds, and the parity dichotomy

Lemma 3. *Let $n \geq 0$ and $d, q, q' \geq 1$ be integers.*

- (i) $\lfloor n/(qd) \rfloor = \lfloor \lfloor n/d \rfloor / q \rfloor$.
- (ii) *If $t \geq 1$ is an integer and $\{q, q'\}$ satisfies $q \geq 2, q' \geq 3$ (in either order), then $t - \lfloor t/q \rfloor - \lfloor t/q' \rfloor \geq 1$. The same holds when $q, q' \geq 3$.*

Proof. (i) is the standard identity $\lfloor \lfloor x \rfloor / q \rfloor = \lfloor x/q \rfloor$ with $x = n/d$; both sides count multiples of qd in $[1, n]$. (ii) $t - \lfloor t/q \rfloor - \lfloor t/q' \rfloor \geq t(1 - \frac{1}{q} - \frac{1}{q'}) \geq t(1 - \frac{1}{2} - \frac{1}{3}) = t/6 > 0$; being an integer, it is ≥ 1 . $\square$

Lemma 4 (Two-term Bonferroni for three sets). *For every $n \geq 0$, $B(n) \geq s(n) - P_2(n)$.*

Proof. Each $k \in [1, n]$ divisible by exactly $j \in \{0, 1, 2, 3\}$ of a, b, c contributes $\mathbf{1}[j \geq 1]$ to $B(n)$ and exactly $j - \binom{j}{2}$ to $s(n) - P_2(n)$ (a k divisible by exactly two of a, b, c is divisible by exactly one pairwise lcm; one divisible by all three is divisible by all three pairwise lcms). Since $j - \binom{j}{2} = 0, 1, 1, 0$ for $j = 0, 1, 2, 3$, each right-hand contribution is at most the corresponding left-hand one. $\square$

Dividing Lemma 4 by n and letting $n \rightarrow \infty$ recovers the classical Heilbronn–Rohrbach inequality $\delta(B) \geq \sum_d \frac{1}{d} - \sum_{d < e} \frac{1}{L_{de}}$ (see Remark 14); the point of what follows is that the finite- n form, with a second subtraction absorbed by integrality, is strong enough to reach the sharp constant 2.

Lemma 5 (Ratios from primitivity). *If $x < y$ and $x \nmid y$, with $g = \gcd(x, y)$, then $L_{xy}/x = y/g \geq 3$ and $L_{xy}/y = x/g \geq 2$.*

Proof. $g \mid y$ and $g \leq x < y$, so $g \neq y$. If $g = y/2$ then $y/2 = g \mid x$ and $y/2 \leq x < y$ force $x = y/2$, i.e. $x \mid y$, a contradiction; so g is a divisor of y other than y and $y/2$, whence $g \leq y/3$ and $y/g \geq 3$. For the second bound, $g = x$ would give $x \mid y$; so $g \leq x/2$ and $x/g \geq 2$. $\square$

Applied to the three pairs of a primitive triple $a < b < c$, Lemma 5 gives

$$\begin{aligned} L_{ab}/a &\geq 3, & L_{ac}/a &\geq 3, & L_{bc}/b &\geq 3, \\ L_{ab}/b &\geq 2, & L_{ac}/c &\geq 2, & L_{bc}/c &\geq 2. \end{aligned} \tag{2}$$

Lemma 6 (Parity dichotomy). *Let $\{a, b, c\}$ be a primitive triple in the uncovered zone (1). Then not both $L_{ac}/c = 2$ and $L_{bc}/c = 2$; i.e. at least one of the last two ratios in (2) is ≥ 3 .*

Proof. Suppose $L_{ac}/c = a/\gcd(a, c) = 2$ and $L_{bc}/c = b/\gcd(b, c) = 2$. Then $\gcd(a, c) = a/2$ (so $a/2 \mid c$) and $\gcd(b, c) = b/2$ (so $b/2 \mid c$). Write $c = k \cdot (a/2)$; since $a \nmid c$, k is odd, and $c > a$ gives $k \geq 3$. Likewise $c = \ell \cdot (b/2)$ with $\ell \geq 3$ odd. From $a < b$ we get $k = 2c/a > 2c/b = \ell$, and both are odd, so $k \geq \ell + 2$. Hence, using $1/a = k/(2c)$, $1/b = \ell/(2c)$, $1/c = 2/(2c)$,

$$\frac{1}{a} = \frac{k}{2c} \geq \frac{\ell + 2}{2c} = \frac{\ell}{2c} + \frac{2}{2c} = \frac{1}{b} + \frac{1}{c},$$

contradicting (1). $\square$

Remark 7. Lemma 6 is exactly where the covered/uncovered split earns its keep. Boundary triples such as $\{6, 10, 15\}$ (with $\frac{1}{10} + \frac{1}{15} = \frac{1}{6}$) have $L_{ac}/c = L_{bc}/c = 2$; they satisfy $\frac{1}{b} + \frac{1}{c} = \frac{1}{a}$, lie in the covered zone, and are handled by Proposition 2.

4 The charge decomposition and the main theorem

Lemma 8 (Charge decomposition). *Let $\{a, b, c\}$ be a primitive triple in the uncovered zone (1). Then $s(n) - 2P_2(n) \geq 3$ for every integer $n \geq c$.*

Proof. Put $t_a = \lfloor n/a \rfloor$, $t_b = \lfloor n/b \rfloor$, $t_c = \lfloor n/c \rfloor$; all are ≥ 1 since $n \geq c > b > a$. Define

$$X_a = t_a - \left\lfloor \frac{n}{L_{ab}} \right\rfloor - \left\lfloor \frac{n}{L_{ac}} \right\rfloor, \quad X_b = t_b - \left\lfloor \frac{n}{L_{ab}} \right\rfloor - \left\lfloor \frac{n}{L_{bc}} \right\rfloor, \quad X_c = t_c - \left\lfloor \frac{n}{L_{ac}} \right\rfloor - \left\lfloor \frac{n}{L_{bc}} \right\rfloor.$$

Each pairwise-lcm floor appears in exactly the two X 's of its constituents, so $X_a + X_b + X_c = s(n) - 2P_2(n)$. By Lemma 3(i), $\lfloor n/L_{ab} \rfloor = \lfloor t_a/(L_{ab}/a) \rfloor = \lfloor t_b/(L_{ab}/b) \rfloor$, and similarly for the other lcms. Thus, invoking (2) and, for X_c , the extra ≥ 3 from Lemma 6:

$$\begin{aligned} X_a &= t_a - \lfloor t_a/(L_{ab}/a) \rfloor - \lfloor t_a/(L_{ac}/a) \rfloor, \quad (\geq 3, \geq 3), \\ X_b &= t_b - \lfloor t_b/(L_{ab}/b) \rfloor - \lfloor t_b/(L_{bc}/b) \rfloor, \quad (\geq 2, \geq 3), \\ X_c &= t_c - \lfloor t_c/(L_{ac}/c) \rfloor - \lfloor t_c/(L_{bc}/c) \rfloor, \quad (\geq 2, \geq 3 \text{ in some order, Lem. 6}). \end{aligned}$$

By Lemma 3(ii) (whose hypothesis is symmetric in q, q') each of X_a, X_b, X_c is ≥ 1 , so $s(n) - 2P_2(n) = X_a + X_b + X_c \geq 3$. $\square$

Theorem 9 (Pointwise criterion). *Let $\{a, b, c\}$ be a primitive triple in the uncovered zone (1). Then $2B(n) > nS$ for every integer $n \geq c$.*

Proof. From $2\lfloor x \rfloor = x + \lfloor x \rfloor - \{x\}$ (where $\{x\} = x - \lfloor x \rfloor$ is the fractional part) applied to $x \in \{n/a, n/b, n/c\}$,

$$2s(n) = nS + s(n) - (\{n/a\} + \{n/b\} + \{n/c\}) > nS + s(n) - 3,$$

since each fractional part is < 1 . Combining with Lemmas 4 and 8,

$$2B(n) \geq 2s(n) - 2P_2(n) > nS + (s(n) - 2P_2(n)) - 3 \geq nS. \quad \square$$

Corollary 10. *For a primitive triple in the uncovered zone and all integers $m \geq 1$, $n \geq c$,*

$$\frac{B(m)}{m} \leq S < \frac{2B(n)}{n}.$$

In particular (EP488) holds for the triple.

Proof. $B(m) \leq s(m) \leq mS$ (union bound and $\lfloor x \rfloor \leq x$), while $S < 2B(n)/n$ is Theorem 9. $\square$

The mechanism above is not special to three generators. Writing, for a finite primitive P and $e \in P$, the charge $X_e(n) = \lfloor n/e \rfloor - \sum_{f \in P \setminus \{e\}} \lfloor n/L_{ef} \rfloor$, the same computation yields the following criterion for sets of any size.

Proposition 11 (Charge-positive sets). *Let $P \subset \{2, 3, \dots\}$ be finite and primitive, with $S = \sum_{d \in P} 1/d$, and suppose that for every $e \in P$*

$$\sum_{f \in P \setminus \{e\}} \frac{\gcd(e, f)}{f} < 1. \quad (3)$$

Then $2B(n) > nS$ for all $n \geq \max P$, and hence $B(m)/m \leq S < 2B(n)/n$ for all $m \geq 1$ and $n \geq \max P$.

Proof. Fix $n \geq \max P$ and $e \in P$; put $t_e = \lfloor n/e \rfloor \geq 1$. By Lemma 3(i), $X_e(n) = t_e - \sum_{f \neq e} \lfloor t_e / (L_{ef}/e) \rfloor$, and since $L_{ef}/e = f / \gcd(e, f)$,

$$X_e(n) \geq t_e \left(1 - \sum_{f \neq e} \frac{\gcd(e, f)}{f} \right) > 0$$

by (3); being an integer, $X_e(n) \geq 1$. Each pairwise-lcm floor lies in exactly two charges, so $\sum_{e \in P} X_e(n) = s_P(n) - 2\Pi(n) \geq |P|$, where $s_P(n) = \sum_{d \in P} \lfloor n/d \rfloor$ and $\Pi(n) = \sum_{\{d, e\}} \lfloor n/L_{de} \rfloor$. Bonferroni gives $B(n) \geq s_P(n) - \Pi(n)$, and $2\lfloor x \rfloor = x + \lfloor x \rfloor - \{x\}$ with each $\{n/d\} < 1$ gives $2s_P(n) > nS + s_P(n) - |P|$; combining,

$$2B(n) \geq 2s_P(n) - 2\Pi(n) > nS + (s_P(n) - 2\Pi(n)) - |P| \geq nS. \quad \square$$

For a primitive triple in the uncovered zone, Lemma 6 is exactly what makes (3) hold: the charge sums for a, b, c are at most $\frac{1}{3} + \frac{1}{3}$, $\frac{1}{2} + \frac{1}{3}$, and (by Lemma 6) $\frac{1}{2} + \frac{1}{3}$, all < 1 . Thus Theorem 9 is the triple case of Proposition 11, and Corollary 10 follows from it. Proposition 11 also proves (EP488) for many larger primitive sets—for instance every pairwise-coprime set the reciprocals of whose three smallest elements sum to < 1 , such as $\{2, 5, 7, 9\}$ or $\{3, 4, 5, 7\}$ (the binding constraint (3) is at $e = \max P$)—though it does *not* close $|P| = 4$ (see Remark 16).

Theorem 12. (EP488) *holds for every finite set $A \subset \{2, 3, 4, \dots\}$ whose primitive core P has $|P| \leq 3$. In particular it holds whenever $|A| \leq 3$.*

Proof. Since $B_A = B_P$ and $\max P \leq \max A$, it suffices to prove the inequality for P and all $m > n \geq \max P$. If $|P| = 1$, this is Proposition 1. If $|P| = 2$, then $\frac{1}{a} + \frac{1}{b} < \frac{2}{a}$, so Proposition 2 applies. If $P = \{a, b, c\}$: when $\frac{1}{b} + \frac{1}{c} \leq \frac{1}{a}$, Proposition 2 applies; otherwise Corollary 10 applies. $\square$

5 Remarks

Remark 13 (The statement typo). The printed source [1] defines B as the integers “no one of which is a multiple of any a ” (the *non-multiples*). That reading is inconsistent with Erdős’s own sharpness witness $A = \{a\}$, $n = 2a - 1$, $m = 2a$, which yields a ratio tending to 2 only under the *multiples* reading (for non-multiples the same witness gives a ratio < 1). The multiples version is confirmed by Erdős’s 1966 restatement and by Guy’s [7] E5; this is the version above. The literal non-multiples version is false (e.g. $A = \{2, 3, 5, 7\}$, $n = 9$, $m = 13$ gives $\frac{3}{13} > 2 \cdot \frac{1}{9}$).

Remark 14 (The backbone is classical: Heilbronn–Rohrbach). Lemma 4 is the finite- n , two-term Bonferroni bound for a set of multiples; at the level of natural density it is the inequality of Heilbronn [3] and Rohrbach [4], $\delta(B) \geq \sum_d \frac{1}{d} - \sum_{d < e} \frac{1}{L_{de}}$, the foundational inequality of the theory of sets of multiples (see Halberstam–Roth [5] Ch. V, Hall [6]; generalized by Behrend and by Ahlswede–Khachatrian). We claim *no* novelty for this bound or for “truncated inclusion–exclusion on $\lfloor n/d \rfloor$ ”. The only element we did not locate in that literature is the packaging used here: keeping the bound finitary and using the integrality of each per-generator charge X_e to gain the *second* subtraction $s(n) - 2P_2(n) \geq |P|$, which upgrades the density bound to the sharp constant 2 for $|P| \leq 3$. This is an elementary refinement of a 1937 result, not a new method.

Remark 15 (Comparison with the transport proof). For the three-generator statement, Chojecki [2] decomposes $B_{\{g_1, g_2, g_3\}} = (B_{\{g_1, g_2\}} \setminus B_{\{g_3\}}) \sqcup g_3\mathbb{N}$ and applies a two-generator theorem to $g_3\mathbb{N}$ and a “pair against one forbidden modulus” theorem to the first block. The latter introduces a restricted counting function $F_{a, b | \{c\}}$, splits on the size of n , and reduces to a lemma asserting the existence

of at least four “exact-one” points, proved by a nested divisibility case analysis ($2b = ka$, then $k \geq 4$ vs. $k = 3$, etc.). That lemma is left as `sorry` in the accompanying Lean files, and Tao (April 2026) gave a family showing the underlying reduction fails for $\min A \geq 3$ —so this is not a cosmetic gap. By contrast, the proof above uses only B itself, no split on n , and no auxiliary two-generator theorem; its single non-routine step is the parity dichotomy of Lemma 6. It is in this sense simpler *for this statement*, with the honest caveat of Remark 16: it is not a simpler road to a more general theorem, and it does not extend.

Remark 16 (Why the method stops at three generators). The pointwise criterion of Theorem 9 is *false* for larger primitive sets, so no direct generalization can succeed. Take $A = \{2p : p \text{ prime}, p \leq 100\}$ (primitive, since $2p \mid 2q \iff p \mid q$; here $|A| = 25$, $\max A = 194$). Then $\delta(B) = \frac{1}{2}(1 - \prod_{p \leq 100}(1 - \frac{1}{p}))$ and $S = \frac{1}{2} \sum_{p \leq 100} \frac{1}{p}$ satisfy $S > 2\delta(B)$ ($0.901 > 0.880$), so $2B(n)/n \rightarrow 2\delta(B) < S$: the criterion $2B(n) > nS$ *fails* for all large n (a direct check: for every $n > 381$). Nonetheless (EP488) holds here, because every element of B is even, so $B(x) \leq \lfloor x/2 \rfloor$ gives $\sup_x B(x)/x \leq \frac{1}{2}$, while a short computation via the scaling recursion $B_{2A'}(x) = B_{A'}(\lfloor x/2 \rfloor)$ (with $A' = \{p \leq 100\}$) gives $\inf_{n \geq \max A} B(n)/n > \frac{1}{4}$. Structurally, for $|P| = 4$ each charge X_i carries *three* lcm-ratios, and the guaranteed values $(2, 3, 3)$ already give $1 - \frac{1}{2} - \frac{1}{3} - \frac{1}{3} < 0$; the charge can be negative. A four-generator argument therefore needs at least three regimes—a charge/union bound where lcm-ratios are large, the “dense half” $B(n) \geq n/2$, and a shared-factor recursion—rather than a single pointwise inequality. Chojecki’s transport machinery, though longer, is aimed at this general case (his Conjecture 4.8 on pair-vs-tail split doubling implies (EP488) for all A); the present argument is deliberately specialized to three generators.

Remark 17 (Formalization). Every step above is finitary and combinatorial. The load-bearing inputs are Lemma 3(ii) (a positive integer bounded below by $t/6$ is ≥ 1) and Lemma 6 (a parity/gcd dichotomy); both are routinely formalizable, and the whole argument can be phrased over $\mathbb{Z}$ (no reals or rationals). In particular it avoids the nested case analysis that the existing formalization leaves unproved, so a `sorry`-free Lean proof of the $|P| \leq 3$ case along these lines appears realistic—which, given that this case is the current stuck point of the public program, seems the most useful thing this note offers.

Acknowledgement of priority. The *statement* of Theorem 12 is Corollary 4.7 of [2] (there proved by other means, modulo the `sorry` noted above); the two-generator and “dense” facts underlying Propositions 1–2 are also present there and in the discussion thread of erdosproblems.com/488. The Bonferroni backbone is the classical Heilbronn–Rohrbach inequality (Remark 14). We claim only a correct, elementary, and formally clean proof of the $|P| \leq 3$ case; a determined referee should still check the finite- n Heilbronn–Rohrbach form and the integrality refinement against Hall [6] and Halberstam–Roth [5] before any novelty claim.

References

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